Customer Analytics Dashboard: Project Documentation

Skyrovix Virtual Internship | Data Analyst track | Prepared by: Karan Tormal

Reporting period: 1 Dec 2010 to 9 Dec 2011

1. Project overview

This project delivers an interactive Power BI dashboard that analyses customer behaviour: how many customers buy, how often they return, where they are located, and how revenue is concentrated across customers. The dashboard contains KPI cards, slicers for Country, Segment and Order count, and four analytical charts with a Top 10 customer view.

2. Data source

The organisation did not provide a separate customer master file. The analysis therefore uses the public **Online Retail** transactional dataset (OnlineRetail.csv, 541,909 rows, 8 columns). Each row is one product line on an invoice. Customer-level figures are derived by grouping these transactions by CustomerID.

Column	Description
InvoiceNo	Invoice number (starts with C for cancellations)
StockCode / Description	Product code and product name
Quantity / UnitPrice	Units sold and price per unit (GBP)
InvoiceDate	Date and time of the transaction
CustomerID	Customer identifier (blank for some rows)
Country	Customer's country

3. Data preparation workflow

The cleaning was done in Python (pandas). The steps were applied in this order:

- Loaded the CSV with UTF-8 BOM handling, so the first column name is read correctly.
- Trimmed spaces and converted Description to upper case; trimmed Country.
- Removed exact duplicate rows.
- Removed rows with blank Description or blank CustomerID (customer-level analysis needs a customer).
- Removed cancelled and returned invoices (InvoiceNo starting with 'C').
- Removed rows with zero or negative Quantity or UnitPrice.
- Removed non-product stock codes: POST (postage), M (manual), C2 (carriage), DOT, BANK CHARGES, PADS.
- Created Revenue = Quantity x UnitPrice and InvoiceMonth (yyyy-mm).
- Built the customer summary table (one row per customer) from the cleaned transactions.

Cleaning step	Rows removed	Rows left
Raw data	-	541,909
Duplicate rows	5,268	536,641

Blank Description / CustomerID	135,037	401,604
Cancelled invoices (C)	8,872	392,732
Zero or negative Quantity / Price	40	392,692
Non-product stock codes	1,542	391,150

Final cleaned dataset: **391,150 rows**, 18,402 invoices, 4,334 customers, 37 countries, total revenue **GBP 8,737,227.64**.

4. Customer summary table

Each row represents one customer (4,334 rows). Columns: CustomerID, Country, FirstPurchase, LastPurchase, Orders (distinct invoices), TotalRevenue, AvgOrderValue, Lifespan_days, Segment (Low / Medium / High, split into three equal groups by revenue), RepeatCustomer (TRUE if Orders > 1), and OrderBucket (1, 2, 3-4, 5+).

5. KPI definitions

KPI	Definition	Value
Total Revenue	Sum of Revenue (Quantity x UnitPrice) on cleaned data	GBP 8,737K
Total Quantity	Sum of units sold	5,141,784
Total Orders	Distinct InvoiceNo	18,402
Total Customers	Distinct CustomerID	4,334
Average Order Value	Total Revenue / Total Orders	GBP 474.80
Repeat Customer %	Customers with Orders > 1 / Total Customers	65.3%
Avg Orders per Customer	Average of Orders in customer summary	4.25
Avg Revenue per Customer	Average of TotalRevenue in customer summary	GBP 2,016

6. Dashboard contents

Visual	Type	What it shows / how to use
KPI cards (5)	Card	Total Revenue, Total Quantity, Total Orders, Total Customer, Average Order
Country slicer	Slicer	Filters every visual by customer country
Orders slicer	Range slider	Filters customers by number of orders (1 to 206)
Segment slicer	Slicer	Filters by High, Medium or Low revenue segment
Top 10 Customers by Revenue	Bar chart (Top N)	The 10 customers with the highest revenue
Revenue by Customer Segment	Donut chart	Share of revenue from High, Medium and Low segments
Customers by Country	Bar chart	Number of customers per country
Monthly Revenue Trend	Line chart	Revenue per month, Dec 2010 to Dec 2011 (chronological)
Customers by Order Frequency	Column chart	Customers grouped by 1, 2, 3-4 and 5+ orders

7. Key findings and insights

- **Repeat customers are strong:** 65.3% of customers ordered more than once, and the average customer places 4.25 orders. Loyalty programmes could increase this further.
- **Revenue is concentrated:** the High segment (1,445 customers, about one-third) generates GBP 7.41M, or 84.85% of total revenue. The top 10 customers account for 17.4% of revenue.
- **One-time buyers are a large group:** 1,505 customers (about 35%) placed only one order. A follow-up campaign after the first purchase could convert this group.
- **Customers are mainly in the UK:** 3,916 of 4,334 customers (about 90%) are in the United Kingdom, followed by Germany, France and Spain. International growth is an opportunity.
- **Seasonal peak:** revenue grew strongly from August 2011 and peaked in November 2011 (about GBP 1.14M), which suggests pre-Christmas demand.

8. Testing

- KPI totals were checked against pandas output: 391,150 rows, 18,402 orders, 4,334 customers, GBP 8,737,227.64 revenue.
- Each slicer was selected and cleared to confirm that cards and charts update, and that values return to the full totals on reset.
- Top N filter on the customer chart was verified to show exactly 10 customers.
- Cross-filtering was checked by clicking chart elements and clearing the selection before reviewing totals.

9. Limitations and notes

- The data is the Online Retail dataset, used as a proxy because no customer master file was provided.
- Rows with a blank CustomerID (about 25% of raw rows) are excluded, so customer KPIs cover identified customers only.
- December 2011 is partial (data ends on 9 Dec 2011), so the last monthly point is lower than the trend suggests.
- Some customers have very high revenue from few orders (for example, customer 16446: GBP 168K from 2 orders). These are treated as valid but flagged as possible bulk orders.
- The country value 'Unspecified' exists in the source data and is kept as recorded.

10. Files submitted

- Customer_Analytics_Dashboard.pbix: Power BI dashboard file
- Customer_Analytics_Dashboard_Documentation.pdf: this document
- Dashboard screenshot (PNG)
- OnlineRetail_clean.csv and customer_summary.csv: cleaned datasets used by the dashboard